

Given:

$$\text{Spectrum} = 45 \text{ MHz.}$$

$$\text{Simplex frequency} = 25 \text{ KHz}$$

$$\begin{aligned} \text{Full Duplex Spectrum} &= \frac{\text{Total Spectrum}}{2 \times \text{Simplex spectrum}} \\ \text{supported voice channels} &= \frac{45000 \text{ KHz}}{25 \text{ KHz}} \rightarrow 1800 \text{ voice channels.} \end{aligned}$$

2. Given: voice Channels = 560.

$$\text{Reuse cell cluster} = 7.$$

$$\text{Channels in a single cell} = \frac{\text{voice channel}}{\text{Reuse cell cluster}} \rightarrow \frac{560}{7} = 80$$

3. Given: Carrier spacing = 200 KHz.

$$\text{No of carriers} = 100.$$

$$\text{time slot} = 8.$$

$$\text{Total simultaneous voice channel} = \frac{\text{No of carriers}}{\text{time slot}} \Rightarrow \frac{100}{8} = 12.5$$

$$12.5 \approx 12 \text{ simultaneous voice channel.}$$

4. Given: Base station radiates = 80 W.

$$\text{Handset Transmission} = 4 \text{ W.}$$

$$\text{Decibel-Based power control ratio} = 10 \cdot \log_{10} \left(\frac{80}{4} \right) \Rightarrow 13.01 \approx 13.$$

5. Carrier to interference ratio = Signal - interference.

$$= -60 \text{ dBm} - (-92 \text{ dBm}) = 32 \text{ dBm.}$$

$$\text{Channels in a single cell} = \frac{\text{Total spectrum}}{\text{Spectrum per channel}} \Rightarrow \frac{42000}{300} \rightarrow 140.$$

$$= \frac{140}{7 (\text{Reuse cell cluster})} \Rightarrow 20 \text{ channels.}$$

7. Power of transmitter = 64 W

$$\text{Received Power} = \frac{64 \text{ W}}{4^2} \Rightarrow 4 \text{ W is the received power}$$

8. Transmit power should be scaled up by 4x because inverse square law (?) is doubled. $2^2 = 4$
9. Total channel count = (cluster size) $\times$ (voice channel) = 60 channel.
10. Doppler shift = $\frac{f \times v}{c} \Rightarrow \frac{27.77 \times 900 \times 10^6}{3 \times 10^8} \Rightarrow 83.31 \text{ Hz}$
 where f is frequency.
 v is the velocity in m/s.
 c is speed of light.
11. Duration of single slot = $\therefore \frac{5}{8} \Rightarrow 0.625 \text{ sec}$
12. Carrier frequency = $\frac{\text{Total spectrum}}{\text{Fix carrier spacing}} = \frac{24000}{200} \Rightarrow 120 \text{ carriers}$
13. Power received = $10 \cdot \log_{10}(8000 \text{ mW}) - 33 \text{ dB}$.
 $39.031 - 33 = 6.031 \sim 6 \text{ dB}$.
14. $P(2.5) = 120 + \log_{10}\left(\frac{2.5}{2}\right) \Rightarrow 120.096$.
 So, if we approximate it we get 123 dB.
15. Total handling capacity = cluster size $\times$ no of calls.
 $= 6 \times 250 \Rightarrow 1500 \text{ calls}$
16. 3 minutes = 180 seconds.
 Calls active at a single time = $\frac{4000}{100} \times 180 = 720000$
17. Total Bandwidth = channel frequency $\times$ Total Channel
 $= 25 \text{ KHz} \times 150 \Rightarrow 3750 \text{ Kbps} \Rightarrow 3.75 \text{ Mbps}$
18. Due to guard band. 30 adjacent band - 1 = 29.
 Spectrum of the band = 40 KHz.
 Wasted Bandwidth = $29 \times 40 \Rightarrow 1160 \text{ KHz}$.

Speed of the vehicle = 90 km/hr.

Diameter of the region = $1.5 \times 2 = 3$ km.

Handoffs in an hour = $\frac{90}{30} = 30$ handoffs

20. Total handling capacity = users $\times$ centers $\times$ Base Stations
 $250 \times 8 \times 40 = 80000$

21. Area = 400 sq km.

Radius = 400 m (or) 0.4 km.

Rollout cells = $\frac{400}{(0.4)^2} = 2500$ cells

22. Channels to be served = $\frac{6 \times 1000 \text{ KHz}}{30 \text{ KHz}} = 200$ channels

23. Erlangs = subscribers $\times$ calls per hour $\times$ average call duration
 $= \frac{1200 \times 4 \times 2.5}{60} = \frac{12000}{60} \Rightarrow 200$ traffic.

24. Maximum users the band can service = $\frac{40000}{400} = 100$ K

25. Total capacity = users $\times$ networks.
 $1200 \times 600 = 720000$

26. Paging rounds = $\frac{200}{4} = 50$. [cell network
Paging scheme]

27. Calls per minute = 40,000
Time for each call = 3 minutes.

fraction of the ordered traffic = $\frac{(40000 \times 3) - 100000}{40000 \times 3} = 0.1667$

28. Subscribers roaming = Total subscribers $\times$ percentage
 $6000000 \times 0.15 = 900000$

29. Percentage of call capacity utilized = $\frac{\text{load}}{\text{Capacity}} \Rightarrow \frac{9000}{12000} = 0.75 \sim 75\%$

30. Channels land in a cell = $\frac{\text{Total Network}}{\text{channel per network} \times \text{cell clusters}}$
 $= \frac{48000}{200} = 60$

31. Channels support = $\frac{\text{spacing allocation}}{\text{GSM deployment}} \Rightarrow \frac{30000}{200} = 150 \text{ channels}$

32. Successfully handed off calls = $0.99 \times 2500 = 2475$
 (handover attempt) calls

33. Path loss = $10 \log_{10}(12000 \text{ mW}) - 45 \text{ dB}$

$40.79 - 45 = -4.2081 \text{ dB}$
 Cluster's overall capacity = $7 \times 900 = 6300$

35. Individual slot length = $\frac{\text{Frame Span}}{\text{equal slot}} = 0.6 \text{ sec}$

36. Doppler shift = $\frac{f \times v}{c} \Rightarrow \frac{41.667 \times 1.8 \times 10^6}{3 \times 10^8} = 250.002 \approx 25.00$

37. Simultaneous calls = 250,000
 Calls per minute = 70,000
 Duration of calls = 4 minutes.

offer load = $\frac{(70000 \times 4) - 250000}{280000} \Rightarrow 1.0714$

38. Combined user capacity = Users $\times$ Cluster
 $600 \times 9 = 4500$

39. Calls being dropped every second.
 No of calls = 60000
 Percentage of call drops = 0.4%

$$\text{calls dropped} = \frac{0.4}{100} \times 60000 = 240 \text{ calls.}$$

$$\begin{aligned} \text{Path loss signal} &= T_x (\text{dBm}) - R_x (\text{dBm}) \\ &= 43 - (-81) = 124 \text{ dBm} \end{aligned}$$

$$\begin{aligned} \text{Total geographical area rollout} &= \text{stations} \times \text{base station area} \\ &= 2.5 \times 150 = 375 \text{ km}^2. \end{aligned}$$

$$\begin{aligned} \text{No of Channels in each cell} &= \left(\frac{\text{Total spectrum}}{\text{each cell spectrum}} \right) \div \text{Reuse factor} \\ &= \frac{48000 \text{ KHz}}{200 \text{ KHz}} \Rightarrow \frac{240}{6} \Rightarrow 40 \text{ channels} \end{aligned}$$

$$\begin{aligned} \text{43. Paging cycle for the entire base} &= \text{users per cycle} \times \frac{1}{\text{subscribers}} \\ &= 6000 \times 150000 \\ &= 900000000 \text{ paging cycles.} \\ \frac{\text{Subscribers}}{\text{users per cycle}} &= \frac{150000}{6000} \Rightarrow 25 \text{ cycles.} \end{aligned}$$

$$\begin{aligned} \text{44 Phone call duration} &= 9 \text{ min} \\ \text{hand off at} &= 3 \text{ min} \\ \text{No of Handoffs} &= \frac{9}{3} = 3 \text{ handoffs} \end{aligned}$$

$$\begin{aligned} \text{45 Signal to noise ratio} &= \text{wanted signal} - \text{noise} \\ &= -85 - (-115) \Rightarrow 30 \text{ dBm} \end{aligned}$$

$$\begin{aligned} \text{46. Minimum bandwidth to support all} &= \text{Users} \times \text{data speed} \\ &= 250 \times 150 \text{ kbps} \\ &= 37500 \text{ kbps.} \end{aligned}$$

$$\begin{aligned} \text{47. Erlangs} &= \frac{1500 \times 4 \times 2.5}{60} \left[\frac{\text{Subscriber} \times \text{average call time} \times \text{min}}{\text{length}} \right] \\ &= 250 \text{ total traffic} \end{aligned}$$

48. Total Bandwidth = control channel + signaling.
 $250 \times 12 \text{ KHz} = 3000 \text{ Kbps}$.

49. Total throughput = $1500 \times 600 \Rightarrow 900000 \text{ Kbps}$.
 (connection) $\downarrow$ average Kbps.

50. Rollout Towers = $\frac{\text{Total Area.}}{\text{area } (\pi R^2) \text{ of Tower}} \Rightarrow \frac{390}{3.14 \times (4)^2} = 6.9665 \approx 7 \text{ Towers}$